Circles

The big idea

A **tangent** is a line that touches a circle at exactly **one** point. This chapter proves two facts about tangents that solve a huge number of geometry problems.

- A line that doesn't meet the circle: **non-intersecting**.
- A line meeting it at two points: **secant**.
- A line meeting it at exactly one point: **tangent**. The point is the **point of contact**.

At any point of a circle there is exactly **one** tangent. From a point inside the circle there is no tangent, on the circle one, and outside exactly **two**.

Theorem 1: tangent ⊥ radius

THEOREM 10.1

The tangent at any point of a circle is **perpendicular** to the radius through the point of contact.

Why: every point Q on the tangent other than P lies outside the circle, so OQ > OP. OP is the shortest distance from O to the tangent line, and the shortest distance to a line is the perpendicular.

Theorem 2: equal tangents

THEOREM 10.2

The lengths of the two tangents drawn from an **external point** to a circle are **equal**.

Proof idea: for tangents PA and PB from P, triangles OAP and OBP are right-angled at A and B, have OA = OB (radii) and share OP. So they are congruent (RHS) and PA = PB. The proof also shows OP bisects ∠APB and ∠AOB.

A TANGENT FROM A POINT 25 CM FROM THE CENTRE HAS LENGTH 24 CM. FIND THE RADIUS.

Radius ⊥ tangent, so $r^2 + 24^2 = 25^2 \Rightarrow r^2 = 49 \Rightarrow$ **r = 7 cm**.

A QUADRILATERAL ABCD IS DRAWN TO CIRCUMSCRIBE A CIRCLE. PROVE AB + CD = AD + BC.

Tangents from each vertex are equal: AP = AS, BP = BQ, CR = CQ, DR = DS. Adding: (AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ), i.e. **AB + CD = AD + BC**.

ANGLE BETWEEN TWO TANGENTS

If tangents from P make angle θ, then the angle at the centre between the radii is 180° − θ, because OAPB has two right angles.

Quick recap

- Tangent touches at one point; radius at that point is perpendicular to it.
- Two tangents from an external point are equal in length.
- These two facts plus Pythagoras solve most problems.

Practice questions

1. The number of tangents that can be drawn to a circle from a point outside it is
   (a) 0 (b) 1 (c) 2 (d) infinite
2. A tangent PQ at P meets a line through the centre O at Q. If OQ = 13 cm and OP = 5 cm, then PQ is
   (a) 8 cm (b) 12 cm (c) 18 cm (d) √194 cm
3. If tangents PA and PB from P are inclined at 70°, then ∠AOB is
   (a) 110° (b) 70° (c) 90° (d) 140°
4. The angle between a tangent and the radius at the point of contact is
   (a) 45° (b) 60° (c) 90° (d) 180°
5. Two concentric circles have radii 5 cm and 3 cm. The length of a chord of the larger circle that touches the smaller circle is
   (a) 4 cm (b) 8 cm (c) 6 cm (d) 10 cm

Answers: 1. c 2. b 3. a 4. c 5. b